System Architecture Overview

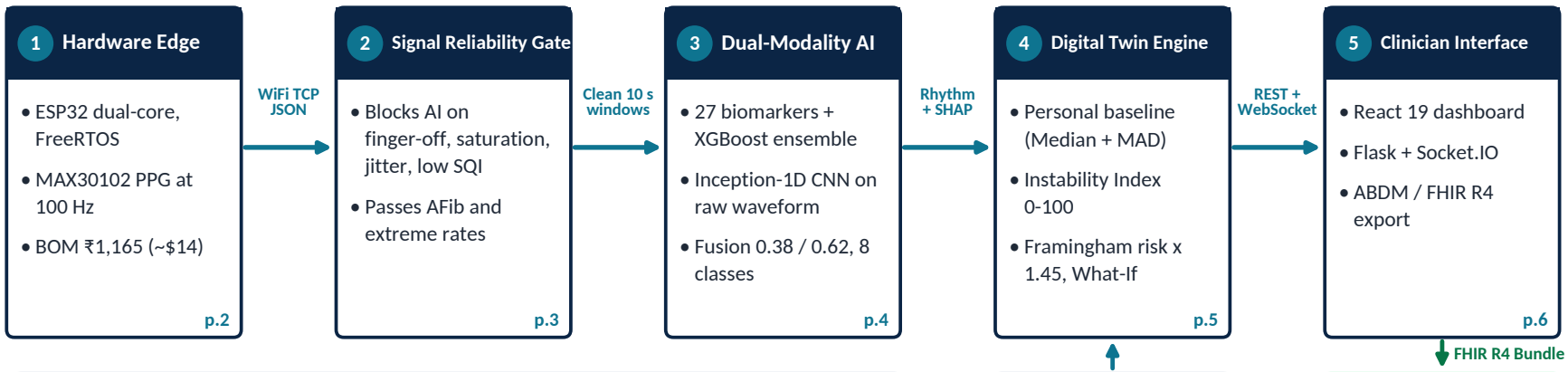
Edge-to-clinic pipeline for personalised cardiovascular screening: read left to right

THE PROBLEM

28% of deaths in India are cardiovascular, and South Asians are hit 10-15 years earlier. Smartwatches (₹41,900+) and bedside monitors are unaffordable; pulse oximeters give one number and no memory.

OUR SOLUTION

A ₹1,165 (~$14) IoT screening node that learns each patient's own baseline, classifies 8 rhythm classes with explainable AI, and projects 10-year risk and What-If treatments for primary-care clinics.



STREAM 2: real-time optical telemetry

100 Hz PPG is captured on the ESP32, gated, cut into 10 s windows (1,000 samples) and analysed by the AI engine.

STREAM 1: patient EHR

Age, sex, SBP, lipids, smoking, diabetes

ABDM / HL7 FHIR R4

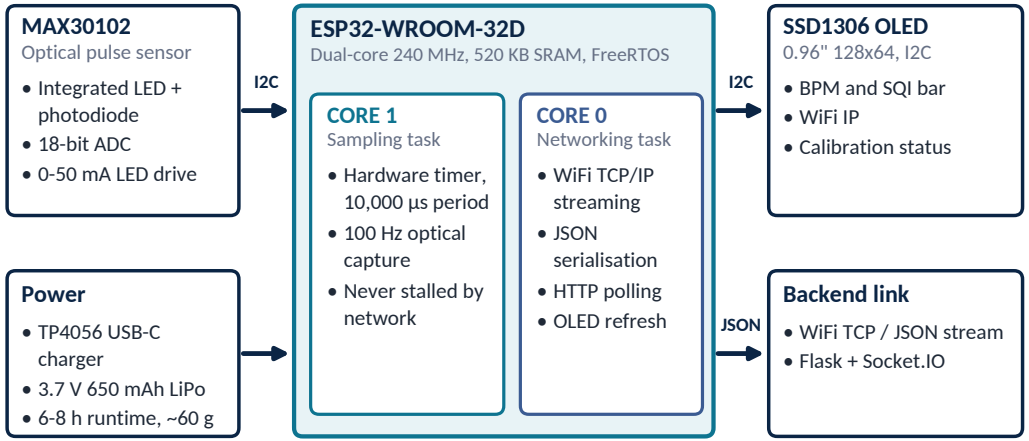
ABHA-linked Bundle export

₹1,165	100 Hz	2,271	53.92%	1.45x	49
hardware BOM (~$14)	PPG sampling	patients, 4,683 windows	Macro-F1, 8 classes	risk recalibration	automated tests passing

Investigational prototype. PPG is not ECG: screening, not diagnosis.

Stage 1: Hardware Edge

ESP32 + MAX30102 node: isochronous 100 Hz acquisition with decoupled networking



Telemetry integrity fields sent with every chunk

device_boot_id
Detects device reboot and config change

seq
Sequential chunk number: detects dropped data

Microsecond timestamps
Enable jitter and drift detection in the gate

Why dual-core: Core 1 owns the 10 ms sampling deadline, so Wi-Fi retries and JSON work on Core 0 can never distort the waveform the AI sees.

Bill of materials

Component	Part	₹
Microcontroller	ESP32-WROOM-32D	480
Optical sensor	MAX30102	220
Display	SSD1306 OLED	140
Power	TP4056 + 650 mAh LiPo	210
Interconnect	Pull-ups, wire, caps	35
Finger clip	3D-printed PLA + foam	80
Total unit BOM	**~$14.07**	**1,165**

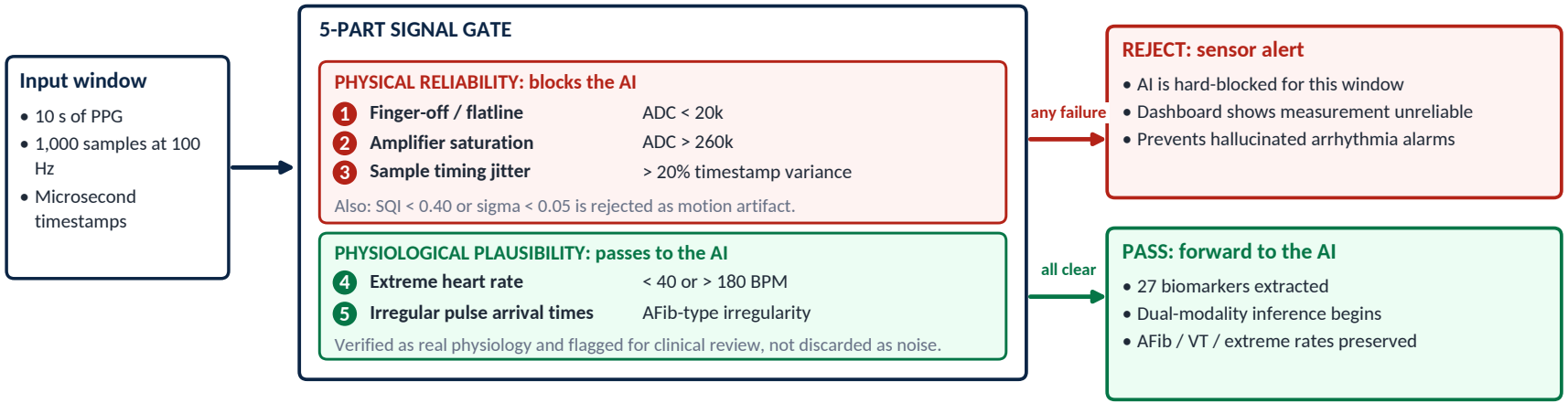
Under ₹950 per unit in a 1,000-unit batch

Cost context

Apple Watch	₹41,900+	Generic baseline
Holter monitor	₹5,000-12,000	Rental, 2-3 day review
Bedside monitor	₹1.2-3.5 lakh	Mains-powered, ICU staff
Pulse oximeter	₹600-1,200	No baseline or memory

Stage 2: Signal Reliability Gate

Decoupled design: physical failures block the AI, physiological irregularities pass through for clinical review



Why decoupling matters

Most pipelines discard anything irregular as noise, which also discards the very arrhythmias the system exists to find. Here, only failures of the measurement itself (finger, amplifier, clock) stop the pipeline. A chaotic but genuine pulse reaches the AI.

Operational validation

100%	0	≤ 2.0 BPM	49
finger-off rejection	persistent false alarms in target validation scenarios	mean absolute error vs reference pulse	automated tests passing

Stage 3: Dual-Modality AI Fusion

Two independent views of the same 10 s window, fused by weights tuned inside nested cross-validation

Clean window
- 10 s
- 1,000 samples
- 100 Hz

MODALITY A: CLASSICAL

Feature extraction: 27 biomarkers
- Time-domain HRV: RMSSD, SDNN, pNN50, pNN20
- Morphology: pulse width, peak-amplitude CV
- Spectral: Welch LF and HF
- Rate and RR statistics, SQI

Classical Super Ensemble
- XGBoost + Random Forest + ExtraTrees
- Soft voting
- Macro-F1 48.00%

Explainability: SHAP TreeExplainer

Per-prediction attribution over the 27 biomarkers shows which features drove each screening result.

x 0.38

MODALITY B: DEEP LEARNING

Raw waveform input
- 1,000 samples, no hand-crafted features
- Direct from the gated PPG window
- Captures waveform shape at four time scales

Inception-1D CNN
- 4 parallel kernels: 3, 7, 15, 31
- Residual skip connections
- 393,224 params, 1.61 MB, ~4.2 ms CPU
- Macro-F1 51.81%

x 0.62

NESTED FUSION

0.38 x Classical + 0.62 x CNN

Weights chosen by inner-fold search; mean across outer folds

8-class output

53.92%

Macro-F1

59.21% accuracy

57.46% bal. acc.

leak-free CV

Target rhythm classes

Normal	Tachycardia	Bradycardia	AFib	Cardiac_Paced	V_Tachycardia	V_Flutter_Fib	Asystole

Stage 4: Digital Twin Engine

Two distinct metrics are tracked and never conflated: acute instability and long-term risk

STREAM 2: Real-Time Instability Index (0-100)

Acute departure from this patient's own resting baseline

1 Calibrate
2-min seated rest; at least 6 valid post-settling 10 s windows

2 Fingerprint
Median and MAD of BPM, RR-CV, RMSSD

3 Score
Robust z = |x - median| / (1.4826 x MAD)

4 Persist
Multi-window temporal consensus before alerting

Guardrails: uncalibrated users get score = null ("Personal baseline required"). Baselines expire after 30 days and are invalidated if device ID or sensor configuration changes.

Instability Index composition

Index component	Signal
Acute rhythm distress	Pr(non-Normal) x 45
Autonomic withdrawal	Penalty when RMSSD < 25 ms
Hemodynamic perturbation	BPM > 105, or severe bradycardia
Smoothing	Exponential moving average, alpha = 0.35

STREAM 1: 10-Year Cardiovascular Risk (%)

Long-term atherosclerotic risk from the clinical record

Inputs
- Age, sex
- SBP
- Cholesterol, HDL-C
- Smoking, diabetes

Framingham Cox PH
- D'Agostino 2008
- Sex-specific coefficients
- 10-year general CVD risk

Recalibration
- x 1.45 South Asian multiplier
- Capped at 100%
- ACC/AHA risk factor

Example: 52-year-old male smoker, SBP 145, TC 220, HDL 38 rises from 16.8% to 24.4%, crossing the 20% high-risk threshold. Ages outside 30-74 are flagged age_extrapolated.

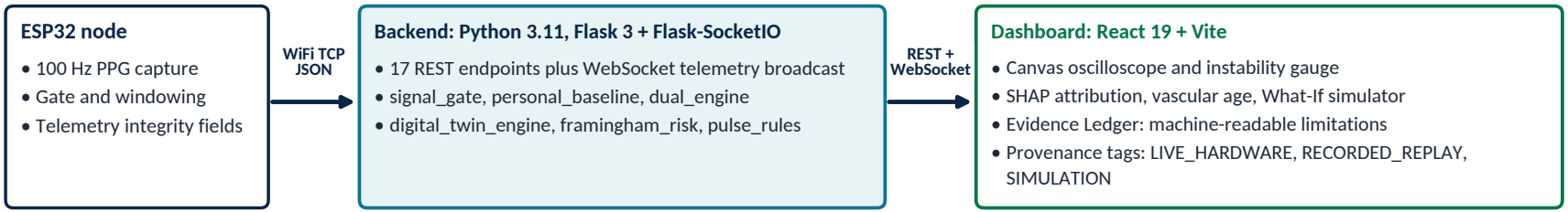
What-If treatment simulator (educational projection)

Intervention	Modelled change	Relative risk shift
Antihypertensives	SBP -15 mmHg	-28% to -34%
High-intensity statin	Total chol. -45 mg/dL	-22% to -28%
Beta-blockade	SBP -8, HR -12 BPM	-15% to -22%
Smoking cessation	Over 5 years	-40% to -50%
Multimodal protocol	**Combined**	**-55% to -68%**

Assumes sustained 5-10 year adherence.

Stage 5: Clinician Interface, Interoperability and Privacy

Real-time dashboard, ABDM / HL7 FHIR R4 export, and an edge-first data boundary



HL7 FHIR R4 mapping, linked to the 14-digit ABHA ID

CardioTwin output	FHIR R4 resource	Coding
Resting heart rate	Observation	LOINC 8867-4
RMSSD (pulse rate variability)	Observation	LOINC 80404-7
Pulse rhythm finding	Observation	SNOMED CT 361137007
Screening pattern (e.g. AFib)	Observation	SNOMED CT 49436004
Personal instability index	Observation	Custom ext-cardiotwin-instability-index
10-year Framingham risk	RiskAssessment	LOINC 79423-0
Hardware node	Device	IEEE 11073, cardiotwin-esp32-01

Every session, baseline and What-If scenario exports as a FHIR Bundle, ready for ABDM and e-Sanjeevani tele-consultation workflows.

Edge-first privacy (DPDP Act 2023 aligned)

1. **Capture**
   Raw 100 Hz waveform lives only in a rolling RAM buffer
2. **Extract**
   27 aggregate biomarkers computed; raw samples purged
3. **Transmit**
   Only aggregate features and departure flags leave the node
4. **Store**
   Pseudonymised handles (PAT001); baselines expire after 30 days

Federated-learning readiness (FedAvg / FedProx) is architectural; multi-centre training is roadmap.

Technology Stack and Standards

Every layer from sensor to clinician, built on open-source tooling

Layer	Technology	Role in CardioTwin
Hardware	ESP32-WROOM-32D (FreeRTOS), MAX30102, SSD1306 OLED, TP4056 + LiPo	Sensing, edge capture, status, power
Firmware	Arduino IDE sketch (.ino), dual-core FreeRTOS tasks	100 Hz sampling on Core 1, WiFi on Core 0
Backend	Python 3.11, Flask 3.0+, Flask-SocketIO 5.3+, Werkzeug 3.0+	17 REST endpoints, live telemetry broadcast
Signal and data	NumPy 1.26+, SciPy 1.13+, Pandas 2.2+, wfdb 4.1+	Signal processing, features, PhysioNet I/O
ML / AI	PyTorch 2.0+, XGBoost 2.0+, scikit-learn 1.4+	Inception-1D CNN, classical ensemble, grouped CV
Explainability	SHAP 0.45+ (TreeExplainer)	Per-prediction biomarker attribution
Training	Google Colab GPU, Joblib model serialisation	Inception-1D training, model artifacts
Frontend	React 19, Vite, Socket.IO client, Lucide; React Native (experimental)	Clinician dashboard, mobile companion
Standards	HL7 FHIR R4, ABDM / ABHA, LOINC, SNOMED CT, DPDP Act 2023	Interoperability and privacy alignment

Open source

MIT License

Tested

49 automated tests (unittest)

Runs without hardware

Simulated scenarios built in